SAVEETHA SCHOOL OF ENGINEERING
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SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

REVIEW 1-PRESENTATION

Predictive Wildfire Spread Simulator with Evacuation Route Optimizer

CSA0915– Java Programming

Presented By:
By:
(192571020)
Lochana Sri R S (192521348)
Nitya Priya P M (192572086)

Guided
S.K.Deepa Arathi
Dr.E.Meganathan

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INTRODUCTION

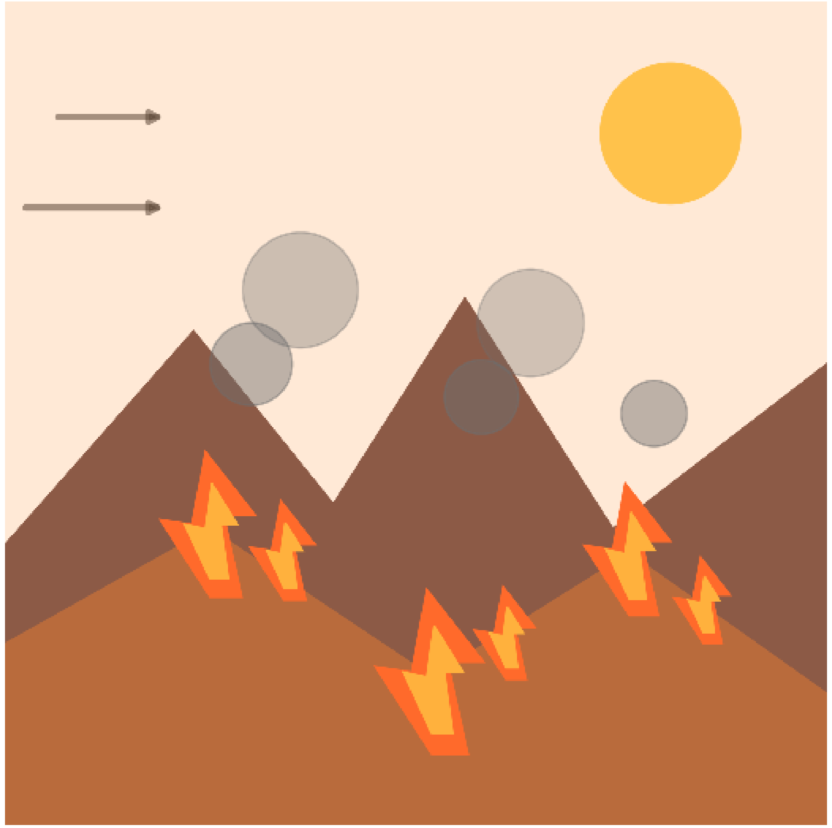
Wildfires are among the most destructive natural disasters, causing loss of life, forests, and property.

Climate change, high temperatures, strong winds, and dry vegetation have increased wildfire frequency and intensity.

Early prediction of wildfire spread is essential to reduce damage and improve emergency response.

FireWatch is an intelligent system that predicts the direction and speed of wildfire spread using environmental data.

The system considers factors such as wind speed, temperature, humidity, vegetation, and terrain.



PROBLEM STATEMENT

Wildfires spread rapidly due to changing weather conditions, making them difficult to predict.

Existing wildfire monitoring systems mainly detect fires after they have already started, limiting response time.

Delayed prediction and evacuation can lead to significant loss of lives, property, forests, and wildlife.

Emergency responders often face challenges in identifying the safest and fastest evacuation routes.

Real-time environmental factors such as wind speed, temperature, humidity, and terrain are not always integrated into decision-making.

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PROPOSED SYSTEM

To develop an intelligent system for predicting wildfire spread.

To analyze wind, temperature, humidity, vegetation, and terrain for better accuracy.

To simulate wildfire spread direction and speed in real time.

To identify the safest and shortest evacuation routes.

To provide early warnings for emergency response teams.

To reduce loss of life, property, and forests through timely evacuation.

To improve disaster management via combined

MODULES

Module 1: Data Collection & Wildfire Prediction

Module 2: Wildfire Spread Simulation

Module 3: Evacuation Route Optimization & Alert System

GEO TAG PHOTO

THANK YOU